Mittal Variety Store: "Driving B2C Success Through Data Driven Solutions Stock Optimization and Sales growth".

A Final report for the BDM capstone Project

Submitted by

Name: **Somiya**

Roll number: **24F2002516**

IITM Online BS Degree Program,

Indian Institute of Technology, Madras,

Tamil Nadu, India, 6000

Content

		Page No.
1.	Executive Summary	2
2.	Detailed Explanation of Analysis Process/Method	3-5
3.	Results and Findings	5-19
4.	Interpretation of Results and Recommendations	19-21
5.	Limitation	21-22
6.	Conclusion	22

Executive Summary

Mittal Variety Store is a neighbourhood-level retail grocery business that caters to daily household needs through a range of grocery items, packaged food, and personal care products. Operating in a highly competitive environment with thin profit margins, the store faces challenges related to inventory control, demand variability, and efficient utilization of working capital. In such a business setting, the application of Business Data Management (BDM) principles becomes essential to support informed decision-making and improve operational efficiency.

This project analyses the sales and inventory performance of Mittal Variety Store using structured data-driven techniques. The study is based on two primary datasets: a sales transaction dataset containing product-level details such as quantity sold, revenue, profit, payment mode, and credit status, and an inventory dataset recording daily opening stock, stock received, and closing stock for each product. As part of the metadata analysis, key variables are identified and descriptive statistics such as mean, median, and standard deviation are used to summarize and understand the underlying characteristics of the data.

The analysis process involves systematic data cleaning, standardization, and integration of sales and inventory records using product identifiers. Multiple analytical methods are applied to examine business performance. **Sales Perfomance analysis** is conducted to evaluate overall trends, category-wise and product-wise performance, and profit contribution. Inventory-focused methods such as **ABC analysis** are used to classify products based on revenue contribution, supporting inventory prioritization. In addition, **Sales–Stock Efficiency Ratio** analysis assesses how effectively inventory is converted into sales at the product level, while **demand forecasting** evaluates stock movement and demand behaviour across different product categories. Payment mode and credit versus non-credit sales analysis are also performed to examine cash flow patterns and financial risk.

Overall, the project demonstrates the practical application of Business Data Management techniques in a small retail context. By combining metadata analysis, descriptive statistics, and method-driven sales and inventory analysis, the study highlights how structured use of data can help identify key performance drivers, improve inventory planning, optimize working capital usage, and support sustainable growth for Mittal Variety Store.

Detailed Explanation of Analysis/Process Method

1. Data Collection and Data Cleaning:

The analysis for this project is based on structured sales and inventory data collected from Mittal Variety Store. Two datasets were used: a sales transaction dataset and an inventory dataset. The sales dataset records daily transactions and includes variables such as date, product identifier, product category, quantity sold, revenue, profit, payment mode, and credit status. The inventory dataset captures daily stock information, including opening stock, stock received, and closing stock for each product.

Before analysis, the datasets were reviewed to understand their structure, variable types, and completeness. This initial assessment helped identify inconsistencies and data quality issues that required preprocessing. Both datasets were maintained in spreadsheet format, enabling systematic organization and further analytical processing.

2. Data Cleaning and Standardization:

A strict data cleaning process was carried out to ensure accuracy and consistency across the sales and inventory datasets of Mittal Variety Store. All variables were first validated for correct data types, with date fields standardized to a uniform format and numerical fields such as quantity sold, selling price, revenue, profit, opening stock, stock received, and closing stock verified for logical and numeric correctness. Missing values were examined carefully, in the sales dataset, missing entries in credit-related fields were observed only for non-credit transactions and were therefore retained to preserve data authenticity. Revenue values were cross-verified using the relationship ***Revenue* = *Quantity Sold* × *Selling Price***, and profit figures were reviewed for logical consistency to ensure internal data integrity.

Product-level standardization was performed using Product_SKU as the primary identifier to maintain consistency between sales and inventory records. Product names and categorical variables such as category, sub-category, payment mode, and credit status were standardized to eliminate duplication and inconsistent labelling. Inventory data was checked for illogical values such as negative stock levels, and outlier values in sales and stock quantities were reviewed to distinguish genuine high-volume transactions from potential data entry errors. These cleaning and standardization steps ensured that the datasets were accurate, consistent, and suitable for advanced analyses such as ABC classification, sales–stock efficiency ratio, and category-level inventory turnover analysis.

3. Data Integration:

To enable combined sales and inventory analysis, both datasets were integrated using product identifiers as the linking key. This integration allowed the relationship between sales performance and inventory availability to be examined at the product and category levels. By combining these datasets, the analysis moved beyond isolated sales reporting and enabled a holistic assessment of inventory efficiency and demand behaviour, which is a central objective of Business Data Management.

4. Data Visualization and Interpretation:

Several analytical methods were applied to address the business objectives of the project. Each method was selected to extract specific insights relevant to retail sales and inventory management.

- **Sales Performance Analysis:** Sales analysis was conducted to evaluate overall business performance as well as category-wise and product-wise trends. Key measures such as total quantity sold, revenue, and profit were used to identify major contributors to sales and profitability. This analysis helped distinguish between high-volume products and high-margin products.
- **ABC Analysis: ABC analysis** was applied to classify products based on their contribution to total revenue. Products were categorized into **A, B, and C** classes, where Category A represents high-revenue products, Category B represents moderate contributors, and Category C represents low-revenue products. This method supports inventory prioritization by highlighting products that require closer monitoring and stricter control.
- **Sales–Stock Efficiency Ratio Analysis**: The Sales–Stock Efficiency Ratio was used to evaluate how effectively inventory was converted into sales at the product level. This ratio compares the quantity sold to the average stock held during the analysis period. Products with higher ratios indicate efficient stock utilization, while lower ratios signal slow-moving inventory. This method assists in identifying inefficiencies in stock allocation.
- **Category-Level Inventory Turnover Analysis:** Inventory turnover analysis was performed at the category level to assess how frequently inventory was sold and replenished within each category. Higher turnover values indicate faster movement of goods and stronger demand, whereas lower turnover suggests slower-moving stock.

This analysis helps in understanding demand patterns across categories and supports category-wise inventory planning.

- **Payment Mode and Credit Sales Analysis:** Transactions were analysed based on payment modes such as cash, digital payments, and credit sales. Credit versus non-credit sales analysis was conducted to assess cash flow behaviour and potential financial risk. This method provides insight into liquidity management and payment preferences of customers.

5. Strategy Formulation and Implementation Planning

To support interpretation, analytical results were presented using appropriate visualizations such as bar charts, line graphs, and summary tables. These visual tools were used to represent sales trends, category contributions, inventory efficiency, and turnover patterns. Visual representation enhances clarity and allows key insights to be communicated effectively.

6. Performance Monitoring and Continuous Improvement

Although the analysis is based on historical data, the methods applied in this project can be extended for continuous monitoring of sales and inventory performance. Regular evaluation of sales trends, inventory efficiency, and turnover metrics can help the business respond proactively to demand changes and maintain operational efficiency over time.

Results & Findings

- ### Sales Perfomance Analysis

Sales data from **January to June 2025** was analysed to understand overall sales performance and demand behaviour. Bar chart was used to represent monthly total sales, highlighting revenue variation across different months. A line graph was further used to observe daily sales trends and identify short-term fluctuations over the analysis period. Product-wise sales performance was examined using quantity sold charts to identify high-demand products, while revenue-based charts were used to highlight top revenue-generating products. In addition, category-wise analysis was conducted by comparing total revenue and profit across categories to understand differences in category performance and profitability. Together, these analyses provide insights into sales trends, product demand, and category-level contribution to overall business performance.

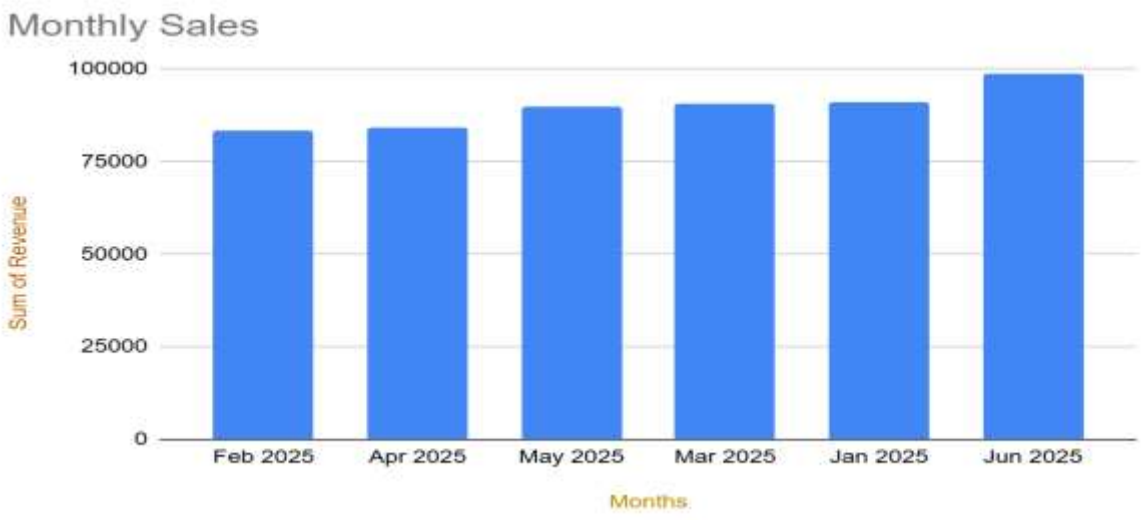

Fig 1: Total Sales Distributed Monthly

Findings:

- Monthly sales show a **gradual increasing trend** over the analysis period.
- Sales are relatively lower during the initial months and **peak in June 2025**, indicating improved demand towards the later period.
- The overall pattern suggests **steady revenue growth** with no extreme month-wise decline.

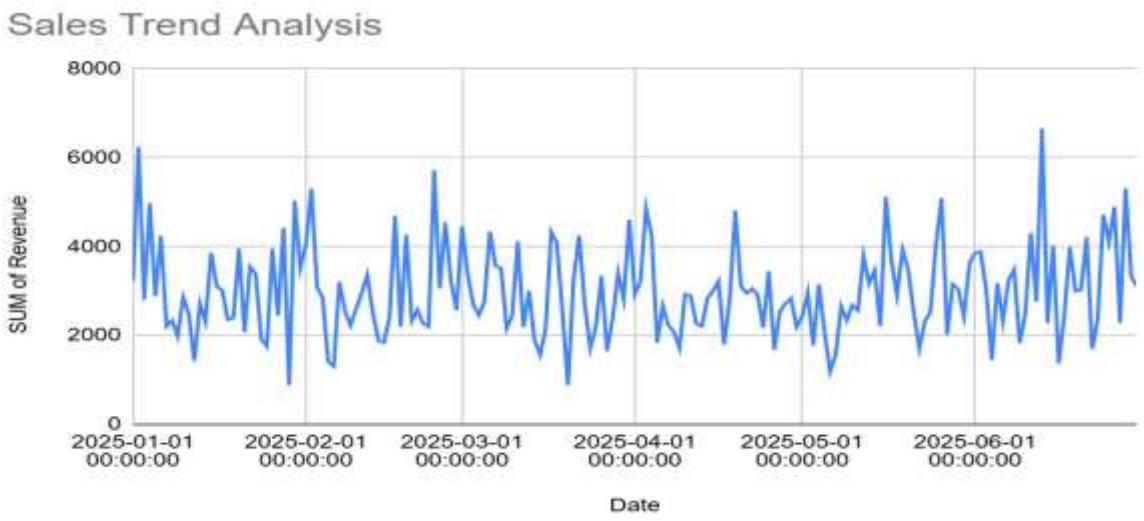

Fig 2 - Total sales trend

Findings:

- Daily sales exhibit **high volatility** with frequent fluctuations throughout the period.
- Multiple short-term peaks and dips indicate **irregular customer purchasing patterns**.

- Despite fluctuations, sales remain consistently active, reflecting **stable business operations over time**

Top Selling Products:

Fig 3: Top 10 Best-Selling Products by Quantity

Findings:

- Products such as **Surf Excel and Lays** record the highest quantity sold, indicating strong and consistent demand.
- Most products exhibit similar sales volumes, suggesting **balanced demand** across frequently used items.
- **High quantity sales** are primarily observed in **daily-use consumer goods**, reflecting routine purchase behaviour.

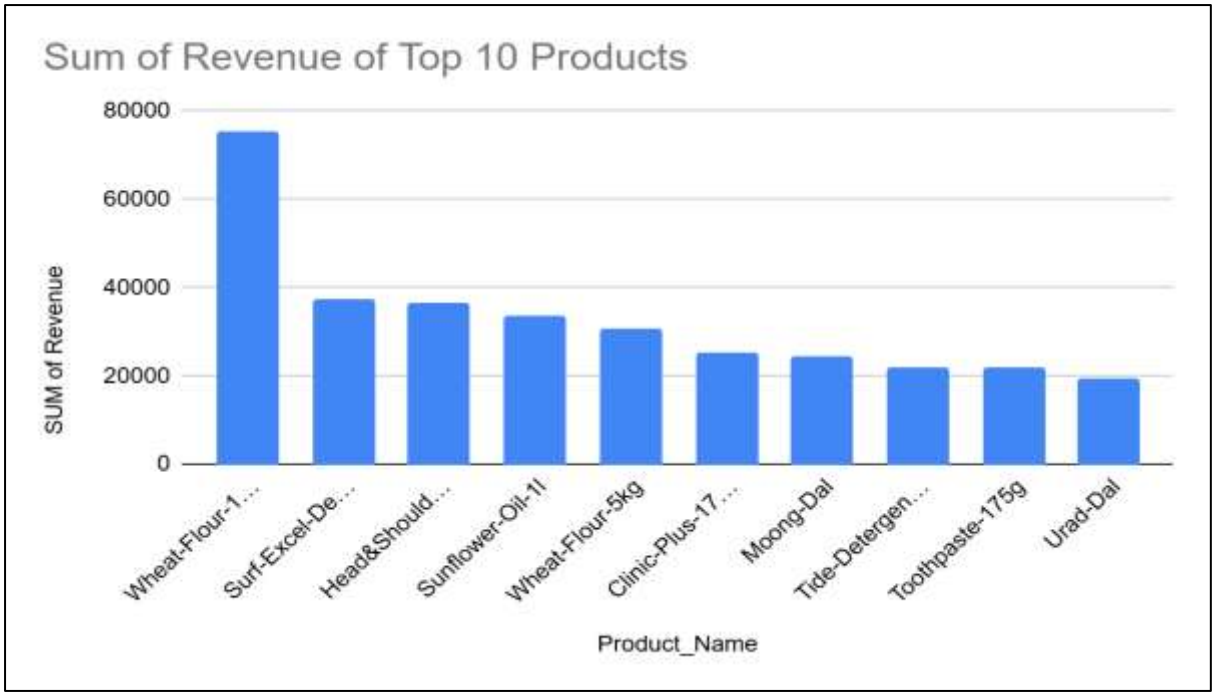

Fig 4: Top 10 Best-Selling Products by Revenue

Findings:

- **Wheat Flour** emerges as the **highest revenue**-generating product among the top 10 products.
- Although some products do not rank highest in quantity sold, they contribute significantly to revenue due to higher unit prices.
- Revenue contribution is concentrated among a few key products, indicating product-level revenue dependence.

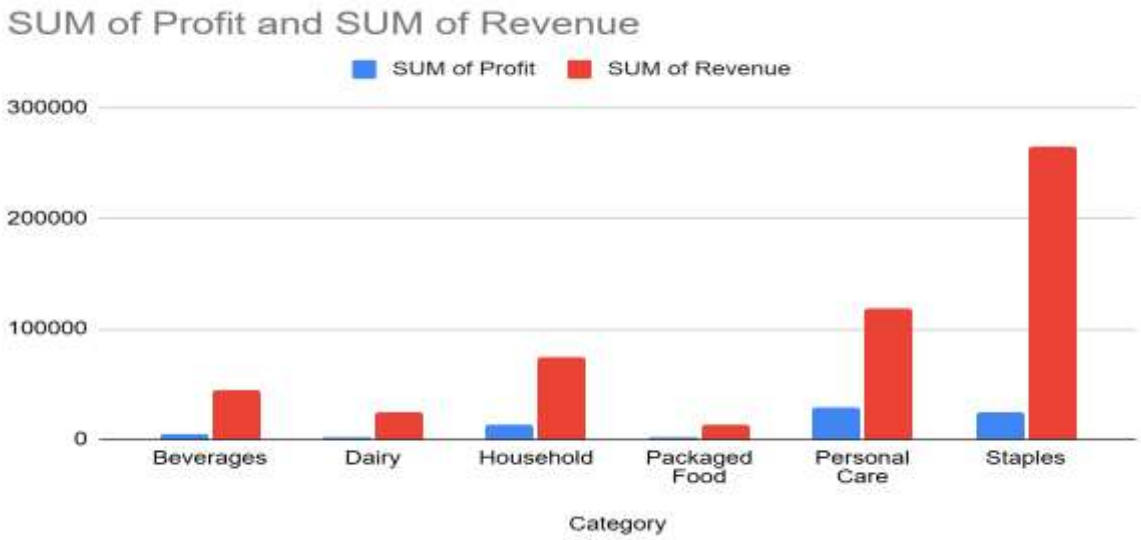

Fig 5: Categorical comparison by Proft and Revenue

Findings:

- The **Staples category** generates the highest revenue as well as profit, making it the most critical category for the business.
- Categories such as **Personal Care and Household** also show strong profit contributions relative to revenue.
- Some categories generate high revenue but comparatively lower profit, highlighting **margin variation across categories**.

- **<u>ABC Analysis</u>**

ABC analysis was performed to classify products based on their contribution to total revenue. Products were arranged in descending order of revenue, and cumulative revenue percentage was calculated to identify the most significant items. Based on this classification, products were grouped into A, B, and C categories. A bar chart along with a cumulative percentage curve was used to visualise revenue contribution across products, while a pie chart was used to show the distribution of products across the three ABC categories.

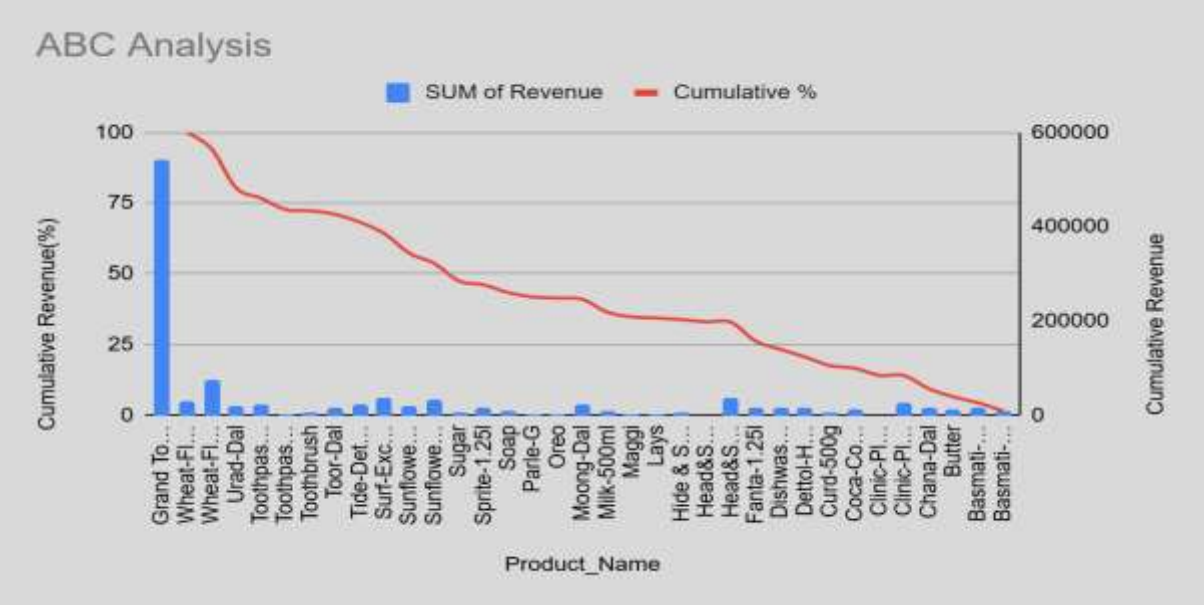

Fig 6: ABC Analysis

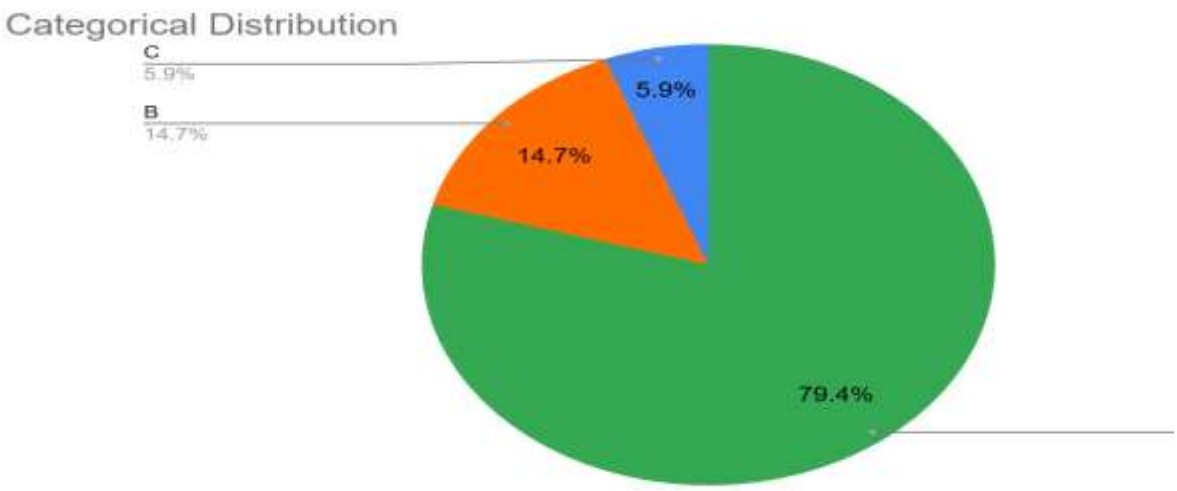

Fig 7: ABC Category Distribution (%)

Findings

- Category **A products contribute a major share of total revenue**, despite representing a smaller portion of the total number of products.
- Category **B products account for a moderate share of revenue** and act as support products that contribute steadily to sales.
- Category **C products contribute the least to overall revenue**, indicating low-impact or slow-moving items.
- The ABC classification highlights **revenue concentration among a limited number of products**, suggesting the need for focused inventory control and monitoring of A-category items.

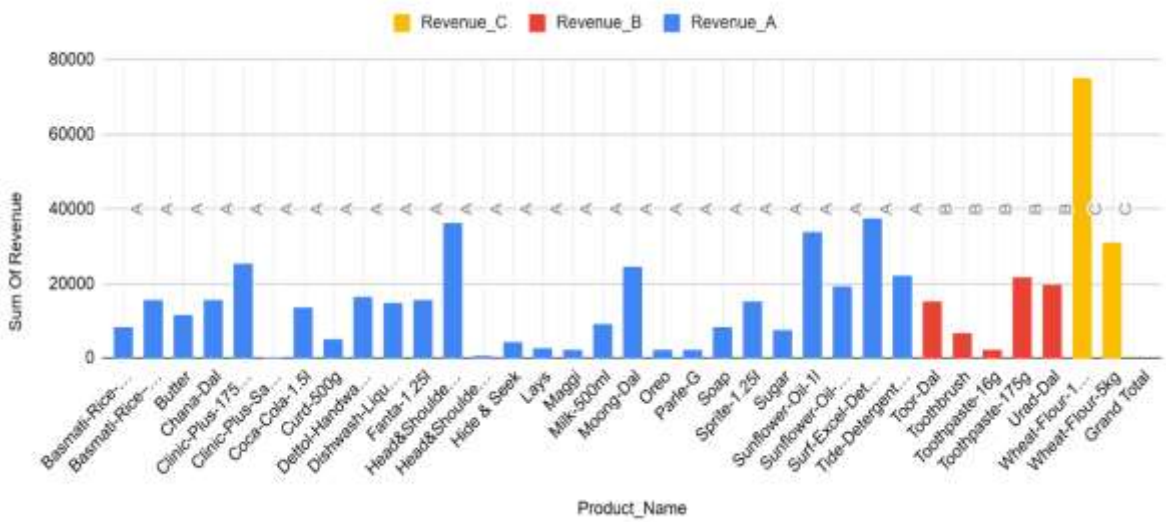

Fig 8: Top Revenue-Contributing Products

Findings: ABC Analysis

- High-revenue essential items like **Wheat Flour, Surf Excel, and Sunflower Oil** fall under Category A and dominate revenue contribution.
- Common grocery items such as **Toor Dal** (pulses) **and personal care products** are classified under Category B with moderate contribution.
- Low-revenue packaged items fall under Category C, indicating limited contribution to overall sales.

- ## Sales-Stock Efficiency Ratio Analysis

Sales–Stock Efficiency Ratio analysis was undertaken to assess the effectiveness of inventory utilisation across individual product SKUs. The efficiency ratio was computed by relating the total quantity sold to the **average stock** held during the analysis period, thereby capturing the extent to which inventory was converted into actual sales. **Product-level efficiency** ratios were visualised using bar charts, while a scatter plot was employed to examine the relationship between average stock levels and quantity sold. Furthermore, products were categorised into fast-moving, medium-efficient, and slow-moving groups based on **predefined efficiency thresholds,** and the distribution of products across these categories was analysed using bar and pie charts.

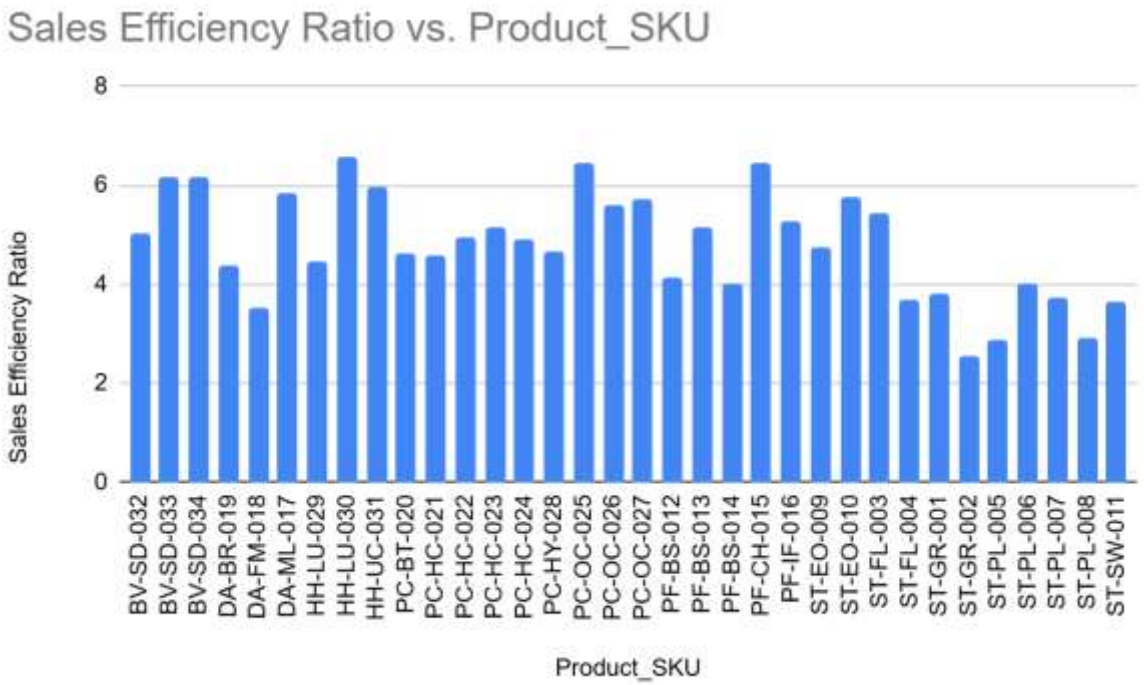

Fig 9: Sales Efficiency Ratio

Findings:

- The Sales–Stock Efficiency Ratio varies considerably across product SKUs, indicating **uneven inventory utilisation** within the product portfolio.
- Several SKUs exhibit **high efficiency ratios (greater than 5)**, classifying them as **fast-moving products** that convert inventory into sales effectively. These products reflect strong demand alignment and optimal stock levels.
- A majority of products fall within the **medium efficiency range (3–5)**, suggesting stable but improvable inventory performance.
- A limited number of SKUs record **efficiency ratios below 3**, categorising them as **slow-moving products** and indicating potential overstocking or weak demand.

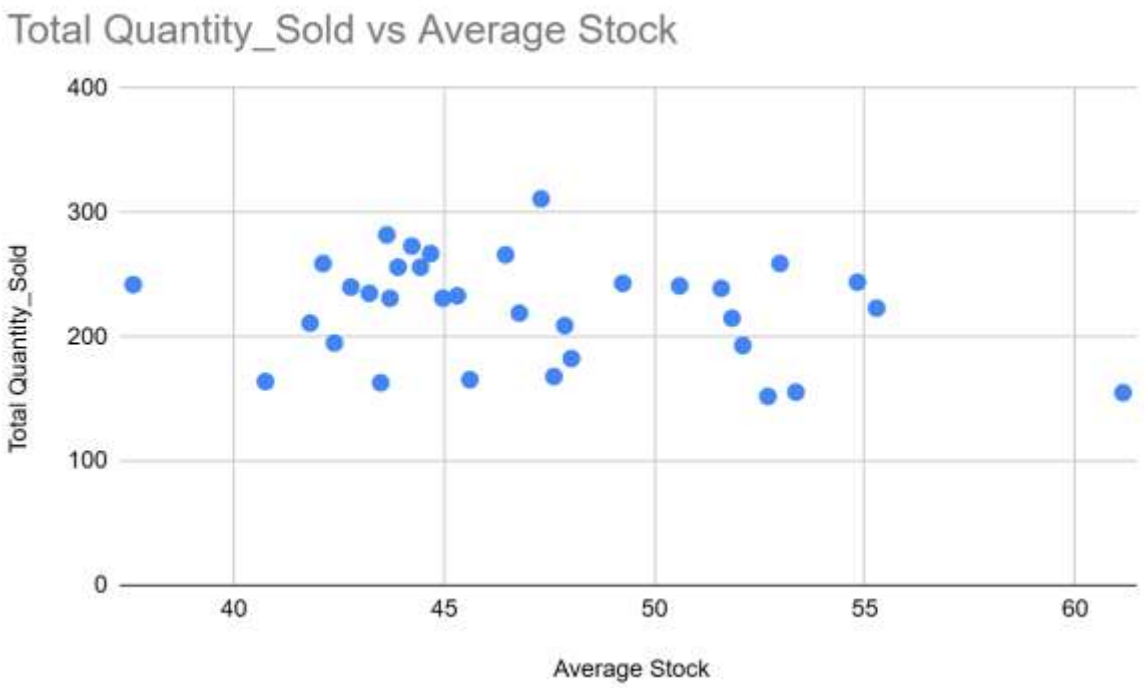

Fig 10: Sales–Stock Relationship Scatter Plot

Findings:

- The scatter plot reveals a **weak correlation between average stock levels and quantity sold**, implying that higher inventory does not automatically translate into higher sales.
- Several products achieve relatively high sales volumes with **moderate average stock**, demonstrating **efficient inventory turnover**.

- Products positioned at higher stock levels with comparatively lower sales volumes indicate **inefficient stock allocation**, increasing holding costs and reducing overall efficiency.
- This analysis highlights the importance of **demand-driven stocking** rather than volume-based stocking.

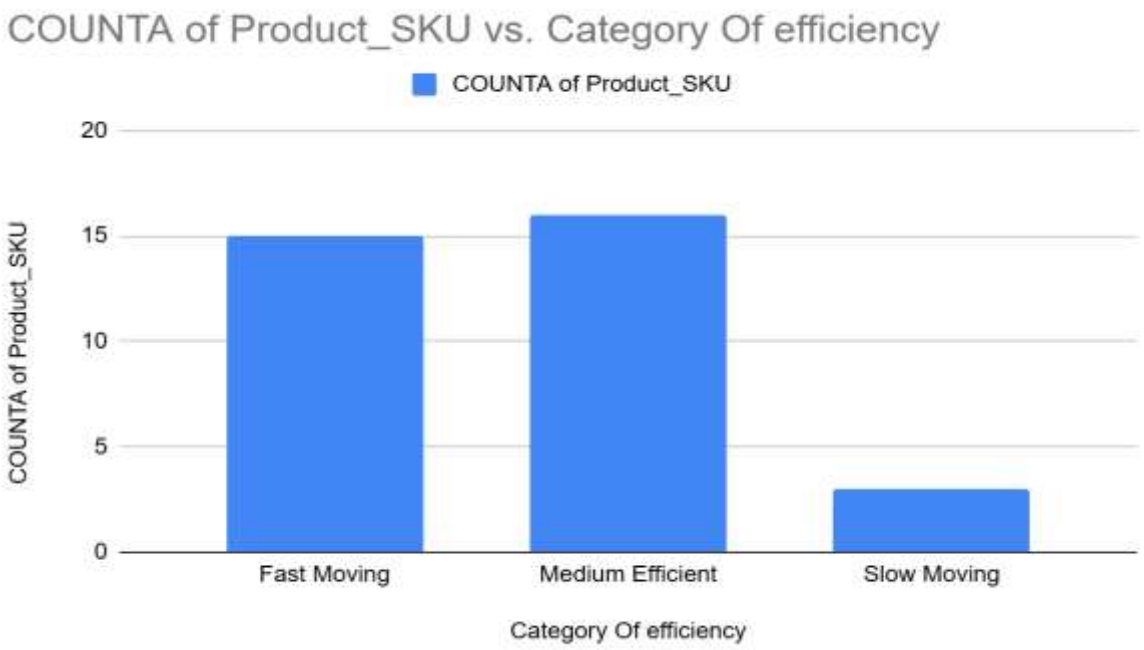

Fig11: Distribution of Products by Sales–Stock Efficiency Category

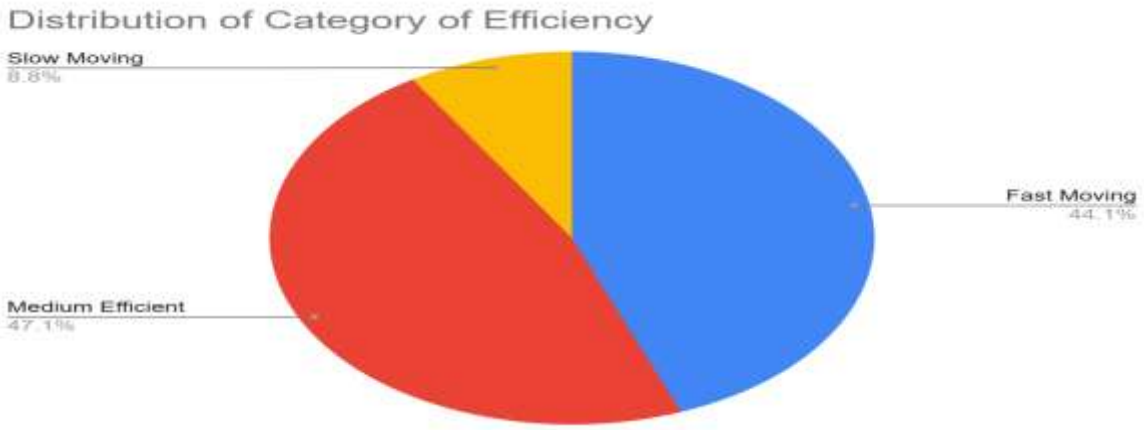

Fig 12: Proportion of Products by Sales–Stock Efficiency Category

Findings:

- The dominance of the **Medium Efficiency category** indicates that most products maintain a **balanced sales-to-stock relationship**, suggesting that inventory levels are neither excessive nor aggressively lean.

- The presence of a **significant Fast-Moving segment** highlights that a subset of products consistently achieves **higher sales conversion per unit of stock**, reflecting strong demand predictability for these SKUs.
- The **limited count of Slow-Moving products** suggests that inefficiencies are **concentrated rather than widespread**, enabling targeted corrective action instead of broad inventory restructuring.
- The close alignment between **percentage distribution (pie chart)** and **absolute SKU count (bar chart)** confirms that the observed efficiency pattern is **structurally consistent** and not driven by a few outlier products.
- Overall, the distribution suggests that **inventory performance is operationally stable**, with improvement potential primarily lying in **re-optimising stock levels of slow-moving items** rather than altering the overall stocking strategy.

- ## Demand Forecasting

Demand forecasting was performed using historical sales data to estimate future demand patterns for top-performing product categories. Weekly and monthly demand aggregation was carried out to capture short-term fluctuations and medium-term trends. A 3-week moving average was applied for weekly forecasting, while a 3-month moving average was used for monthly demand smoothing. This method helped reduce random demand variability and highlight underlying demand trends. The comparison between actual demand and forecasted values enabled the identification of demand stability and volatility across categories.

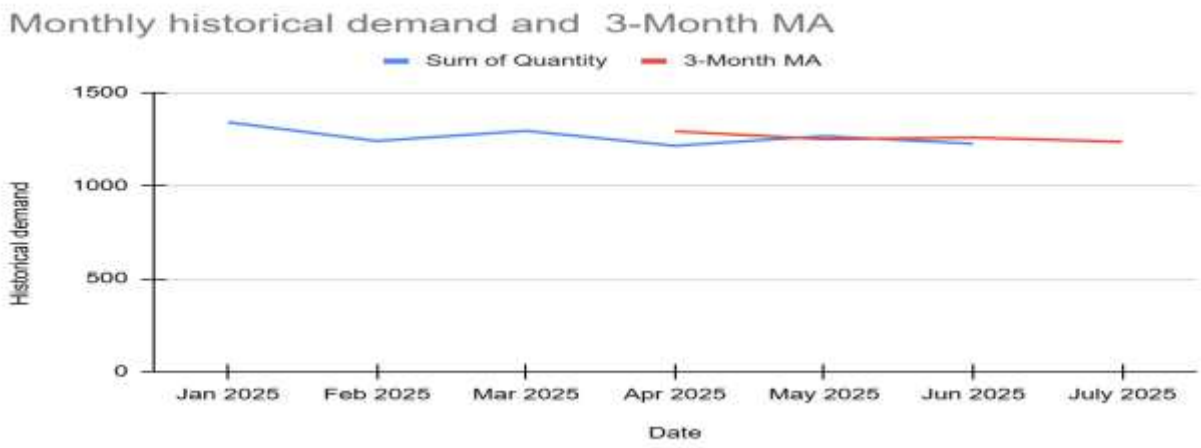

Fig 13: Demand Forecasting of Total sales

Findings

- Monthly demand analysis reveals **gradual demand stabilization** over the observed period.

- The 3-month moving average indicates a **slight downward adjustment in demand**, signalling potential demand saturation or seasonal normalization.
- Actual monthly demand values remain close to the moving average, reflecting **strong alignment between observed and forecasted demand**.
- This trend supports **medium-term planning decisions**, such as procurement cycles and warehouse capacity allocation.
- The analysis confirms that monthly demand forecasting can effectively guide **strategic inventory planning**.

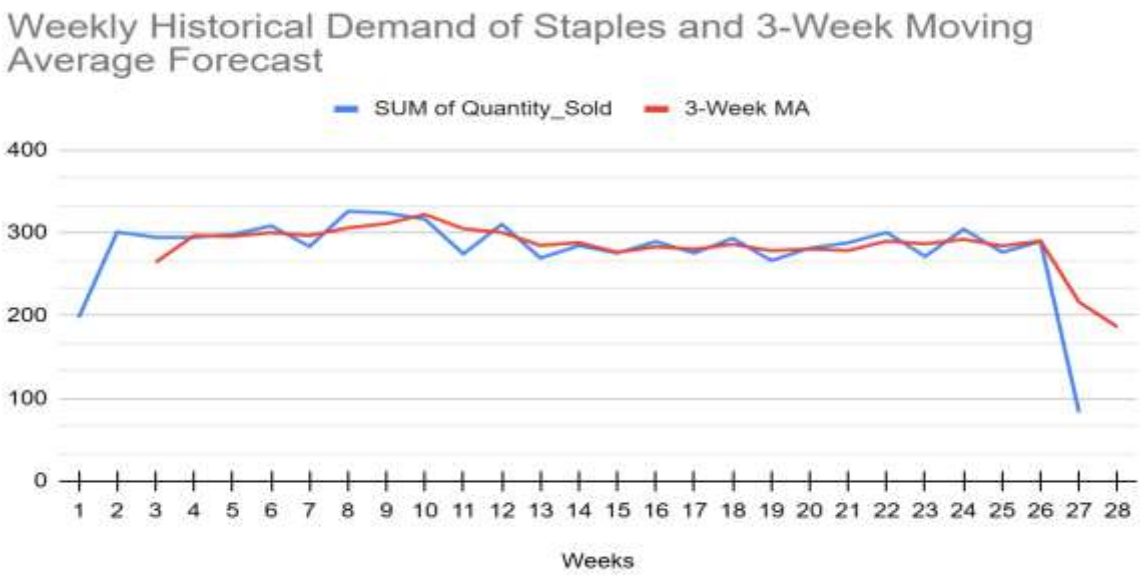

Fig 14:Demand Forecasting of Personal Staples

Findings

- The Staples category exhibits relatively stable weekly demand, with quantities largely fluctuating within a controlled range.
- The 3-week moving average closely follows actual demand, indicating low volatility and predictable consumption patterns.
- Minor demand peaks observed around mid-weeks suggest periodic buying behavior, possibly linked to household restocking cycles.
- The sharp decline in the final week reflects data boundary effects rather than a true demand collapse, reinforcing the importance of excluding incomplete periods in forecasting.
- Overall, the Staples category is highly suitable for forecast-based replenishment, with minimal risk of overstocking.

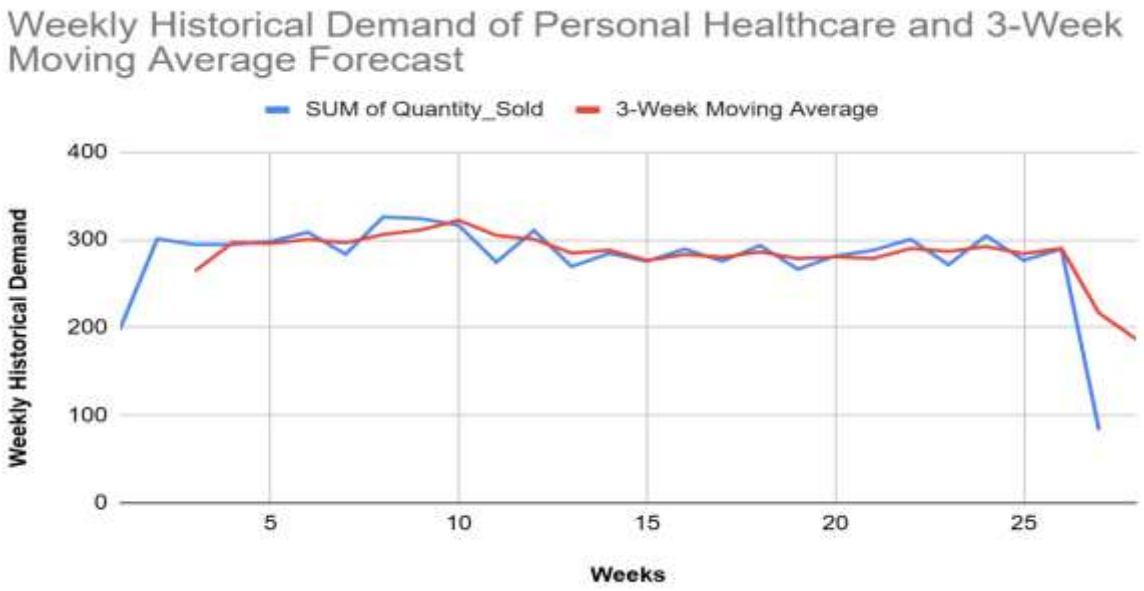

Fig 15: Demand Forecasting of Personal Health Care

Findings

- Personal Healthcare products display moderate demand variability, slightly higher than Staples.
- The moving average smooths short-term spikes effectively, highlighting an underlying steady demand trend.
- Occasional fluctuations suggest event-driven or need-based purchases, such as hygiene or healthcare-related consumption.
- Despite variability, the forecast line remains stable, indicating that short-term forecasting remains reliable for this category.
- This category benefits from buffer stock strategies to accommodate sudden demand surges without causing stockouts**.**
- **Payment Mode and Credit Sales Analysis**

This analysis evaluates the impact of different payment modes and credit transactions on overall revenue performance. Multiple visualizations were used to compare revenue contribution, credit dependency, billing behaviour, and revenue volatility across payment modes. The objective is to assess financial risk exposure, customer payment behaviour, and operational stability from a business owner's perspective. By combining revenue share, trend analysis, average bill value, and variability measures, the analysis provides actionable insights for optimizing payment strategy and managing credit risk.

Fig 16: Payment Mode Wise sales Contribution

Findings

- **Cash and UPI contribute nearly equal** and dominant shares of total revenue, indicating strong dependence on instant payment modes.
- Credit sales form a very small portion of total revenue, suggesting controlled credit exposure.
- The business is primarily liquidity-driven, reducing short-term cash flow risk.

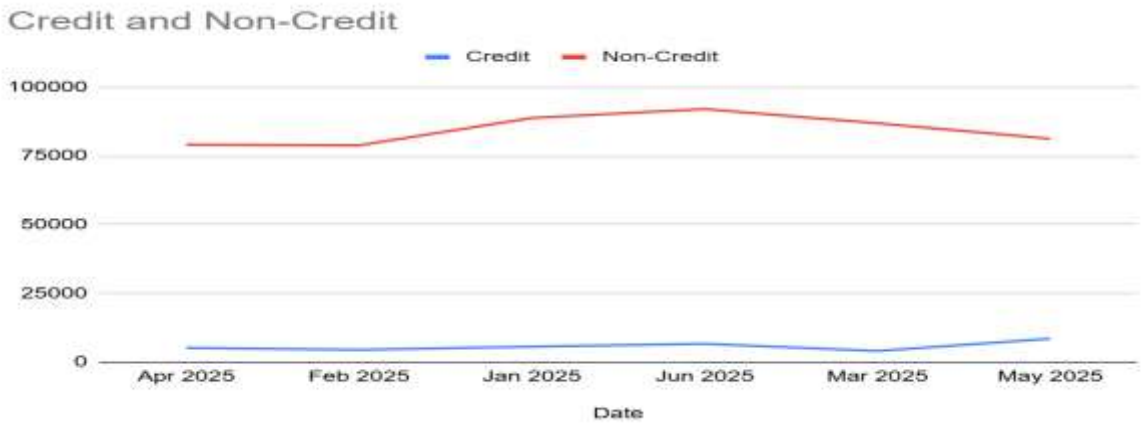

Fig 17: Credit vs Non-Credit Sales Trend

Findings

- Non-credit sales remain consistently high and stable across all months.
- Credit sales show low volume but visible fluctuations, indicating selective and irregular credit usage.
- This trend suggests that credit is used more as an exception rather than a core sales strategy.

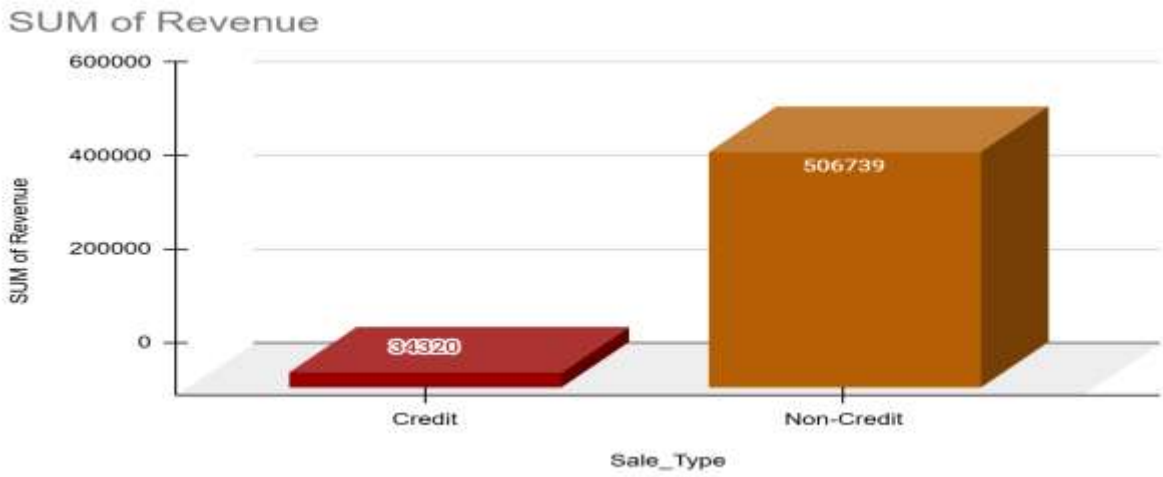

Fig 18: Revenue Contribution by Sale Type

Finding

- Non-credit transactions generate overwhelming majority of revenue, clearly dominating total earnings.
- Credit contributes a minimal revenue share, reinforcing a low-risk sales structure.
- The business is not overly dependent on deferred payments, ensuring financial discipline.

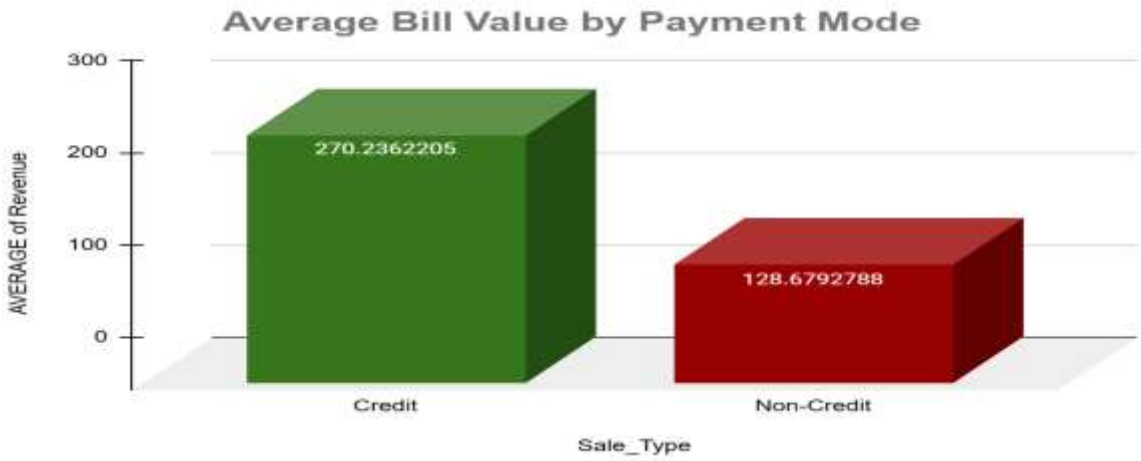

Fig 19: Average Bill Value by Sale Type

Findings

- Credit transactions have a significantly higher average bill value compared to non-credit sales.
- This indicates that customers tend to purchase higher-value baskets when using credit.
- While credit volume is low, each credit transaction carries higher monetary exposure.

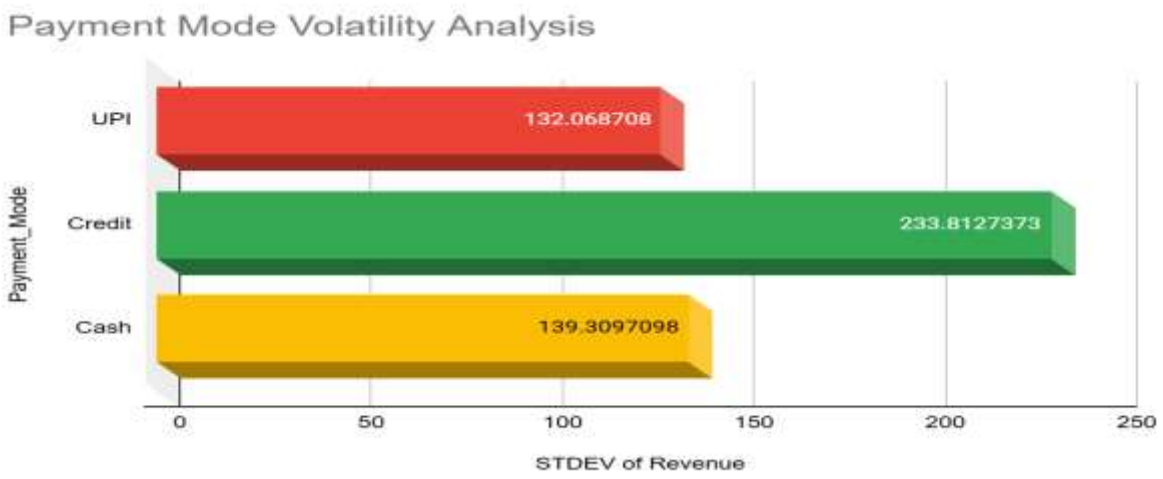

Fig 20: Payment Mode Volatility Analysis

- Credit payments exhibit the **highest revenue volatility**, reflecting irregular usage and higher uncertainty.
- Cash and UPI show **lower and more stable volatility**, indicating predictable daily cash inflows.
- From a risk perspective, credit sales introduce **earnings instability despite low volume**.

Working and Analysis link:
https://docs.google.com/spreadsheets/d/1ekzEBOidlMHaIioYrrjMztPwqjARNEx5vfcuxYBCdYo/edit?usp=sharing

Interpretation of Results and Recommendations

The comprehensive analysis of grocery sales, inventory, demand patterns, and payment behaviour reveals several critical insights into the operational and financial dynamics of the business. These findings provide a data-driven understanding of sales performance, inventory efficiency, demand stability, and credit risk exposure.

- **Sales Performance and Revenue Structure:** The sales analysis indicates that overall revenue is strongly driven by non-credit transactions, with UPI and cash payments contributing the largest share of total sales. This highlights a digitally inclined customer base while maintaining a significant dependence on traditional cash transactions.
- **Sales–Stock Efficiency:** The sales–stock efficiency ratio reveals clear segmentation between fast-moving, medium-moving, and slow-moving products. High-efficiency SKUs demonstrate optimal inventory turnover, whereas low-efficiency products exhibit excess stock holding and weak sales conversion, increasing carrying costs.

- **Demand Forecasting and Trend Stability:** Weekly and monthly demand forecasting using moving average techniques indicates relatively stable demand for staple and personal healthcare categories, with limited short-term volatility. The moving average closely tracks historical demand, validating its suitability for short-term operational forecasting.
- **Credit Sales and Risk Exposure:** Credit transactions form a smaller share of total revenue but exhibit a significantly higher average bill value and revenue volatility. This suggests that while credit customers generate larger transactions, they introduce higher financial risk and revenue uncertainty.
- **Payment Mode Volatility:** Variance and standard deviation analysis across payment modes shows that credit-based sales are the most volatile, followed by cash, while UPI transactions demonstrate comparatively stable revenue behaviour. This positions digital payments as the most predictable and controllable revenue stream.
- **ABC Analysis and Revenue Concentration:** ABC classification shows that a small proportion of products (Category A) accounts for a disproportionately high share of revenue, confirming the Pareto principle. Category B products show moderate contribution, while Category C products contribute marginally despite higher SKU counts, indicating potential inefficiencies in assortment planning.

Recommendations:

Based on the analytical insights derived from the study, the following strategic recommendations are proposed to enhance operational efficiency, financial stability, and decision-making effectiveness:

- **Optimize Category C Inventory:** Rationalize slow-moving Category C products through discontinuation, bundling, or targeted promotions to reduce excess inventory holding costs and improve overall stock efficiency.
- **Adopt Forecast-Driven Inventory Planning:** Integrate demand forecasting outputs into replenishment decisions, especially for high-volume categories, to minimize stockouts and overstocking while improving service levels.
- **Control Credit Exposure:** Introduce credit limits, stricter repayment cycles, or selective credit eligibility for customers to reduce revenue volatility and financial risk associated with credit sales.

- **Promote Digital Payment Adoption:** Encourage UPI-based transactions through incentives or discounts, as they offer stable revenue patterns, faster cash flow realization, and lower operational risk compared to credit sales.
- **Implement Efficiency-Based SKU Monitoring:** Regularly monitor sales–stock efficiency ratios to identify declining SKUs early and take corrective actions, ensuring a dynamic and responsive inventory strategy.

Limitations:

Despite providing valuable insights, the analysis is subject to certain limitations that should be considered while interpreting the results:

- **Limited Time Horizon of Data**: The analysis is based on a relatively short historical period, which may not fully capture long-term seasonal trends, demand cycles, or structural changes in customer behaviour.
- **Simplified Demand Forecasting Techniques:** Demand forecasting was performed using moving average methods, which assume stable demand patterns and may not accurately account for sudden demand shocks, promotions, or external market disruptions.
- **Absence of Customer-Level Credit Risk Data**: Credit sales analysis does not incorporate customer credit history, repayment timelines, or default behaviour, limiting the precision of credit risk assessment.
- **Static Assumptions in Inventory Metrics:** Sales–stock efficiency and inventory analyses assume stable lead times and replenishment cycles, whereas real-world supply chains may experience variability due to supplier or logistics constraints.
- **Limited External Market Considerations:** The study does not account for external factors such as competitor pricing strategies, inflation, seasonal festivals, or macroeconomic conditions, which can significantly influence sales and payment behaviour.
- **Aggregated Product and Category Analysis:** Certain analyses are conducted at category or SKU-aggregated levels, which may mask product-level nuances such as substitution effects or cross-category dependencies.

Conclusion

This project presented a comprehensive data-driven analysis of Mittal Variety Store, focusing on sales performance, inventory efficiency, demand behaviour, and payment mode dynamics

to support informed managerial decision-making. Through structured exploratory and analytical techniques, the study successfully transformed transactional data into actionable business insights.

Sales and revenue analysis revealed a clear dominance of non-credit transactions, indicating strong cash flow discipline and limited dependency on deferred payments. Payment mode analysis highlighted the growing importance of digital transactions (UPI), which contribute significantly to total revenue while exhibiting lower volatility compared to credit sales. In contrast, credit transactions, although contributing a smaller share of revenue, were found to have higher average bill values and greater revenue volatility, signalling elevated financial risk exposure.

Inventory and sales–stock efficiency analysis enabled the classification of products based on their movement and contribution, supporting data-backed inventory rationalization and working capital optimization. The ABC analysis further emphasized the revenue concentration in high-performing product categories, reinforcing the need for focused stock availability and strategic pricing for key items.

Demand forecasting using weekly and monthly moving averages demonstrated relatively stable demand patterns across major categories, validating the feasibility of forecast-driven replenishment planning. The alignment between historical demand and moving averages provides a reliable foundation for reducing stockouts and overstocking risks.

Overall, the analysis establishes that Mittal Variety Store operates with strong financial discipline, efficient inventory utilization, and increasing digital payment adoption, while also identifying areas for improvement in credit risk management and demand forecasting sophistication. By leveraging these insights, the store can enhance operational efficiency, strengthen cash flow stability, and sustain long-term business growth through data-driven strategies.